

M1 — What this block gets you

- ✓ Your SSIS mental model, unchanged— sources → transforms → sinks maps 1:1 onto Google Cloud
- ✓ Where every piece lives in GCP— console, projects, IAM, billing, regions
- ✓ Why file format matters more than file location— CSV vs Parquet sets up this afternoon
- ✓ BigQuery as your target warehouse— serverless, no instance, no patching
- ✓ Hands-on: orient in the console, create your class dataset, then your first load lab

This morning, block by block

09:25–10:10 M1 — Data engineering tasks & components, plus GCP platform orientation

10:10–10:25 Console orientation — star the demo projects, create your class dataset

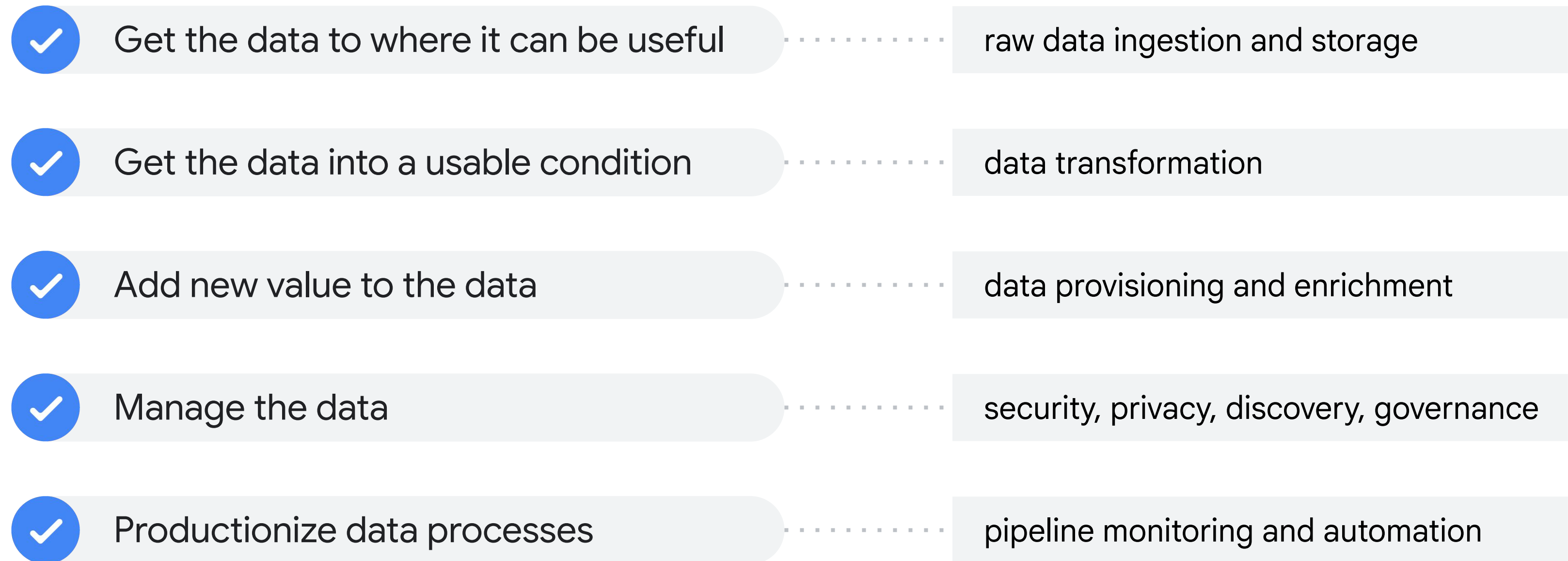
DIN

10:25–10:40 Break

10:40–11:10 Lab: Loading Data into BigQuery — your first win in a new console

LAB

A data engineer builds data pipelines to enable data-driven decisions

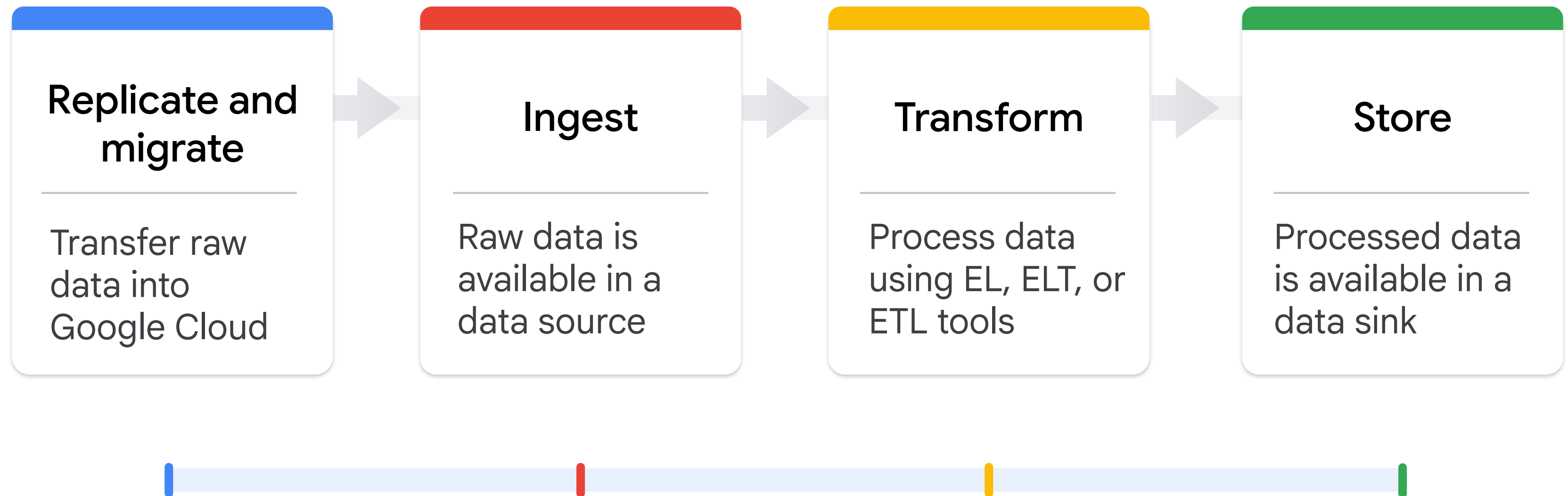


Your SQL Server stack, translated

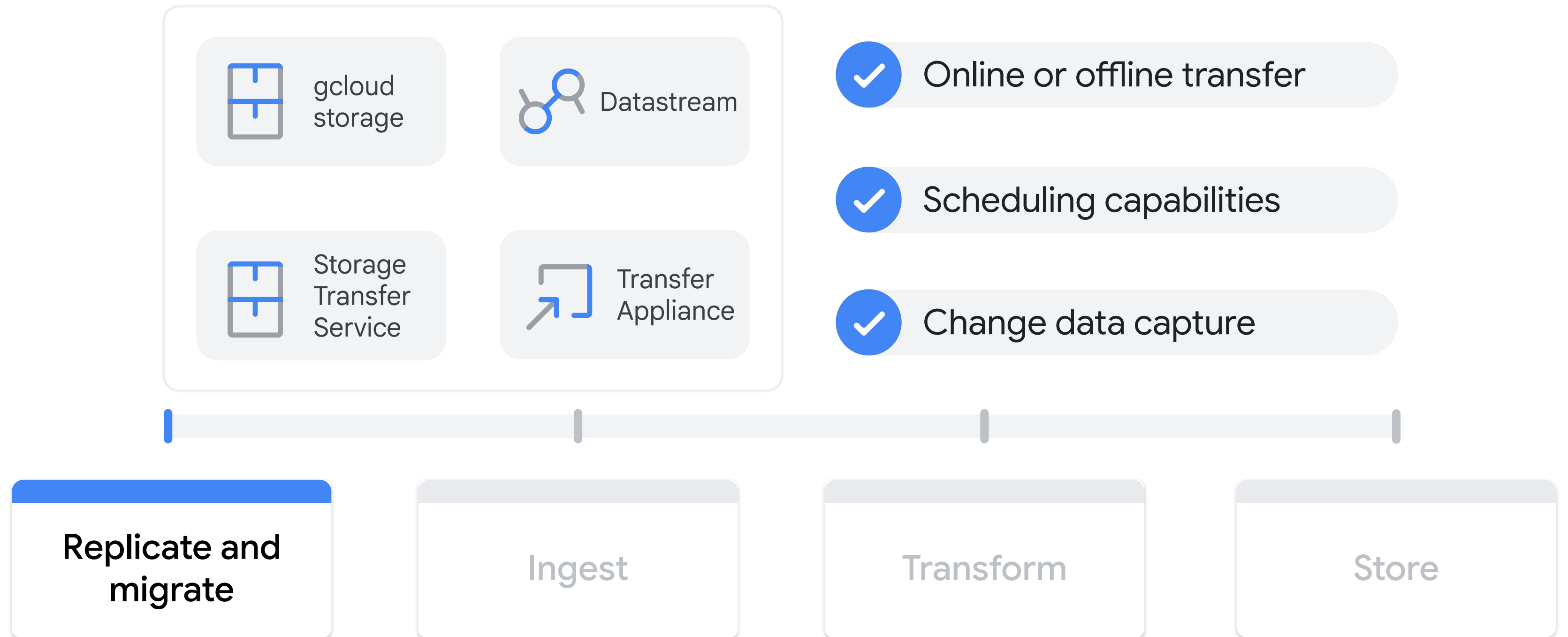
Everything on the next slides is a tool you already own, wearing a Google name.

- SSIS data flow— sources → transforms → sinks, unchanged as a mental model
- SQL Agent jobs— Cloud Scheduler and BigQuery scheduled queries (end of Day 1)
- SSIS packages in git + CI/CD— Dataform repositories and release configs (this afternoon)
- Nightly extract windows— Datastream change data capture, replicating continuously (M2)
- Production SQL Server— stays the system of record while BigQuery becomes the analytics source of truth
- Your data marts → Power BI— Gold marts in BigQuery feed the reports you already have (Day 2)

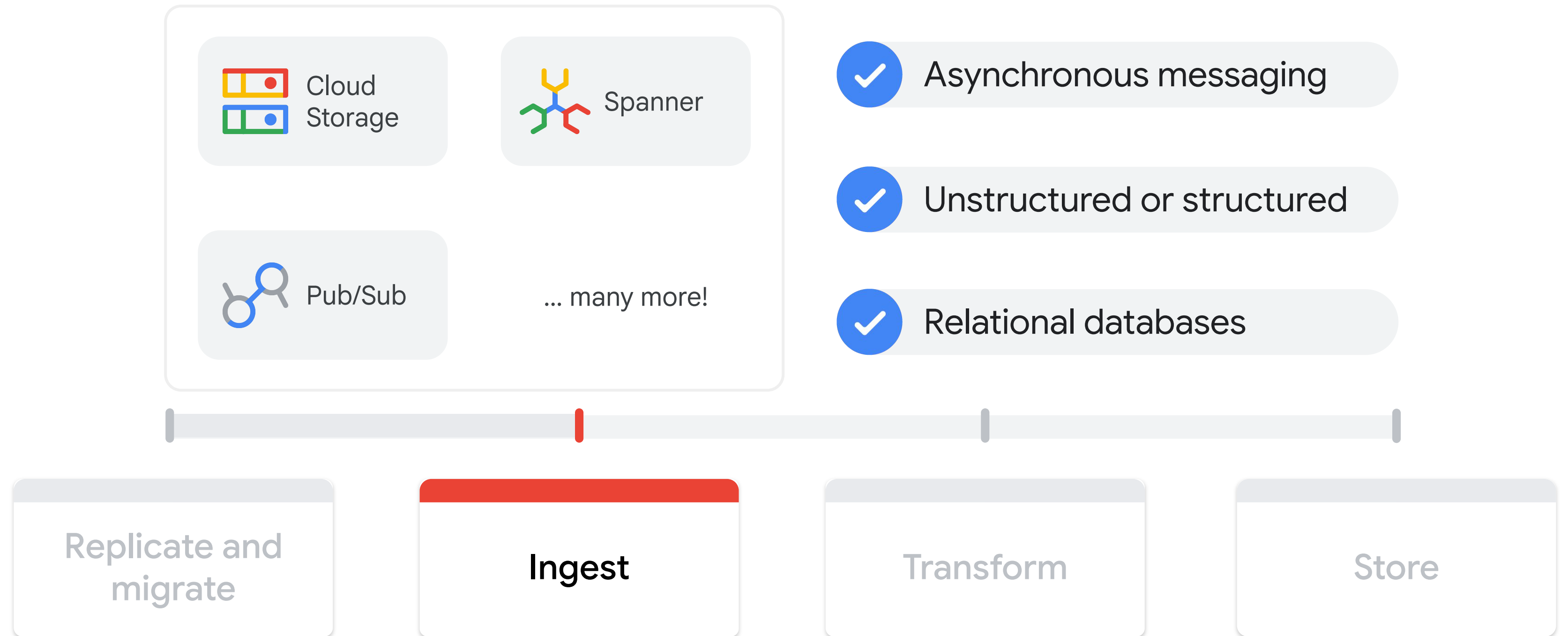
Data engineering tasks evolve around ingesting, transforming, and storing data



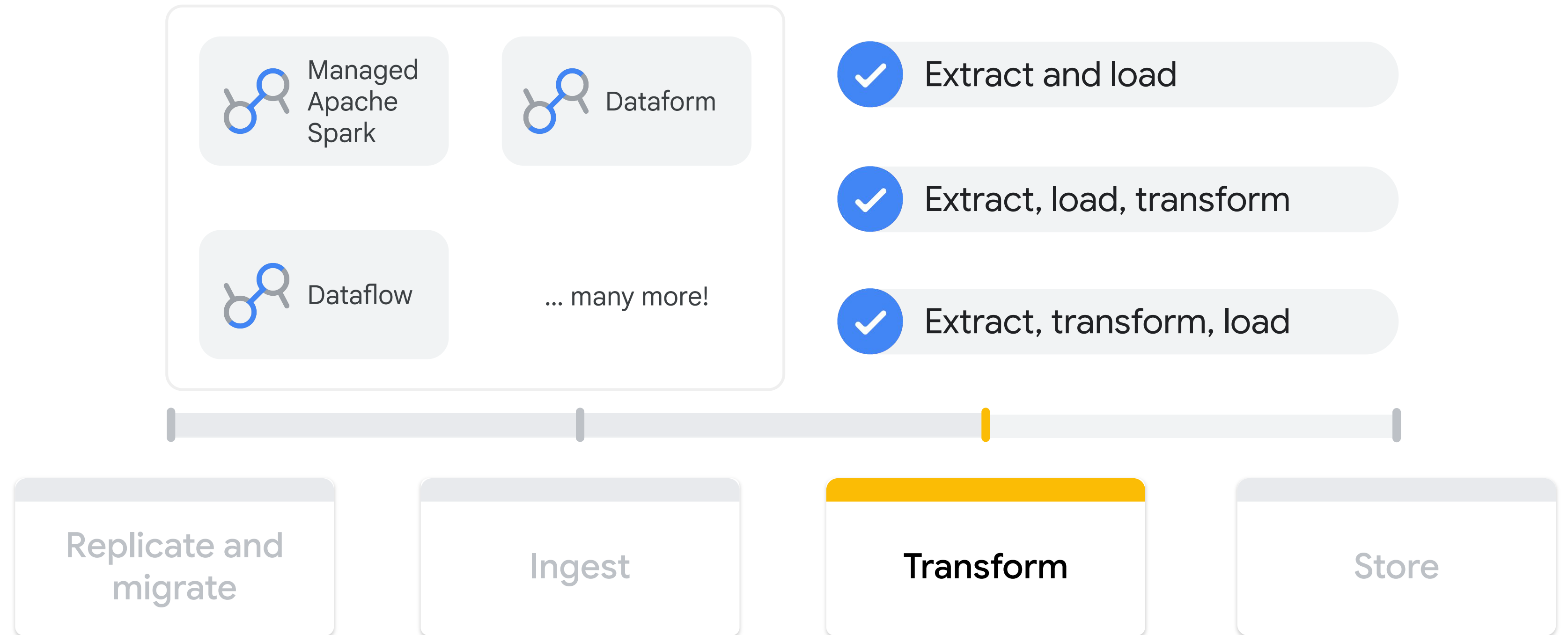
Replication and migration services onboard your data into Google Cloud



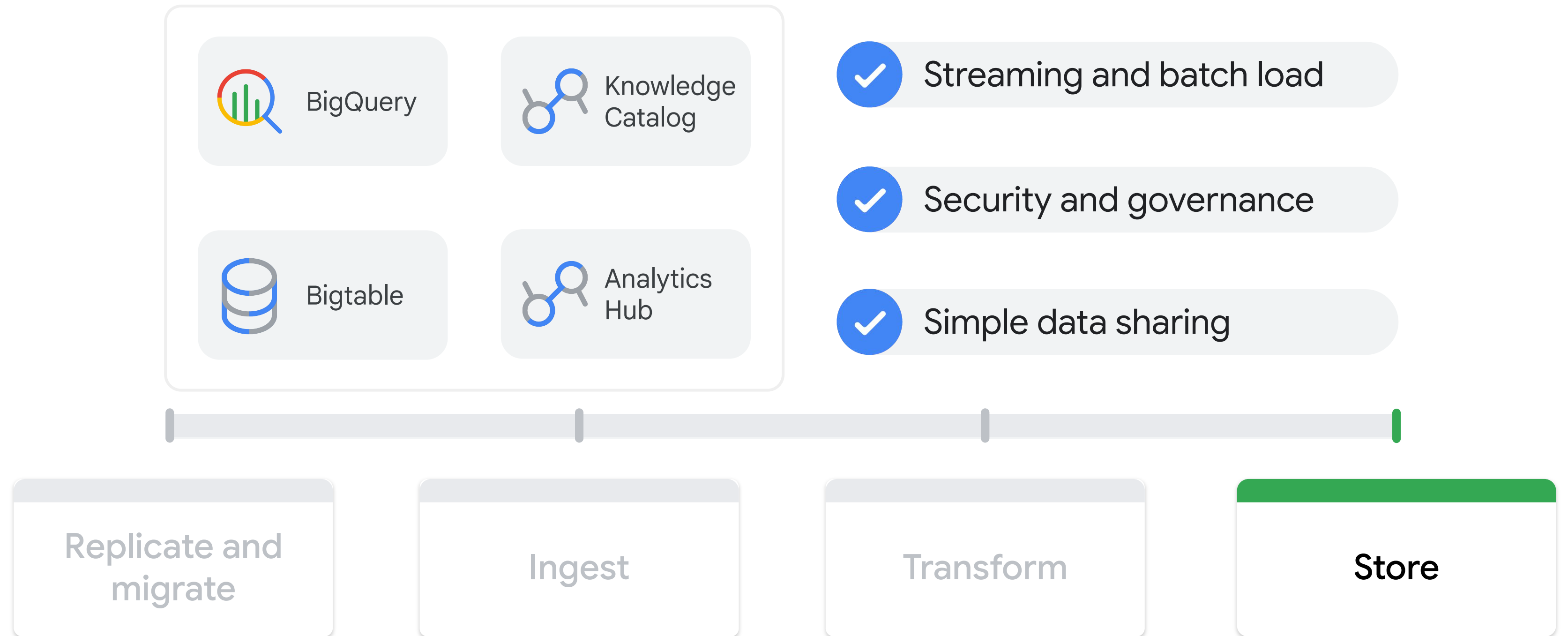
Data sources are the origin point of your raw data on Google Cloud



Transformation services add new value to your data



Data sinks store your processed data on Google Cloud



Data can have different formats

Storage/process data

✓ Data type

✓ Business need

Unstructured



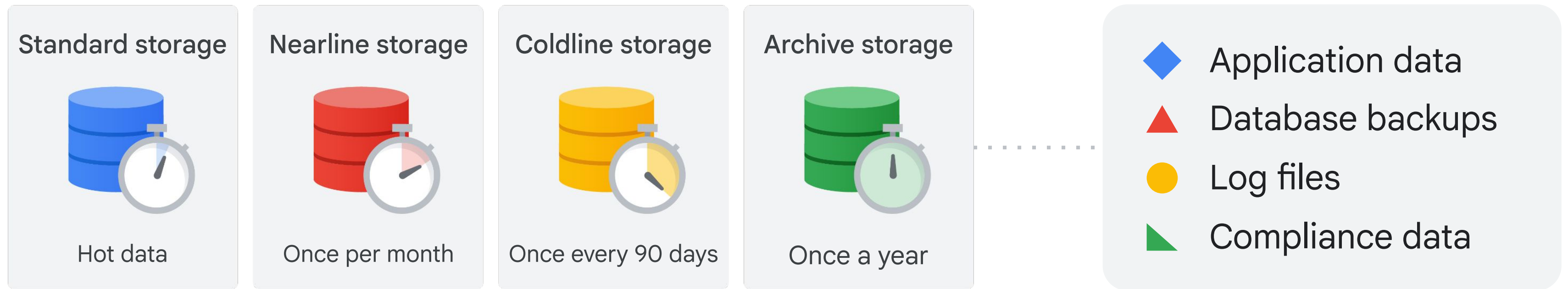
- ◆ Documents
- ▲ Images
- Audio files

Structured



- ◆ Tables
 - └ Rows
 - └ Columns

Cloud Storage holds your unstructured data



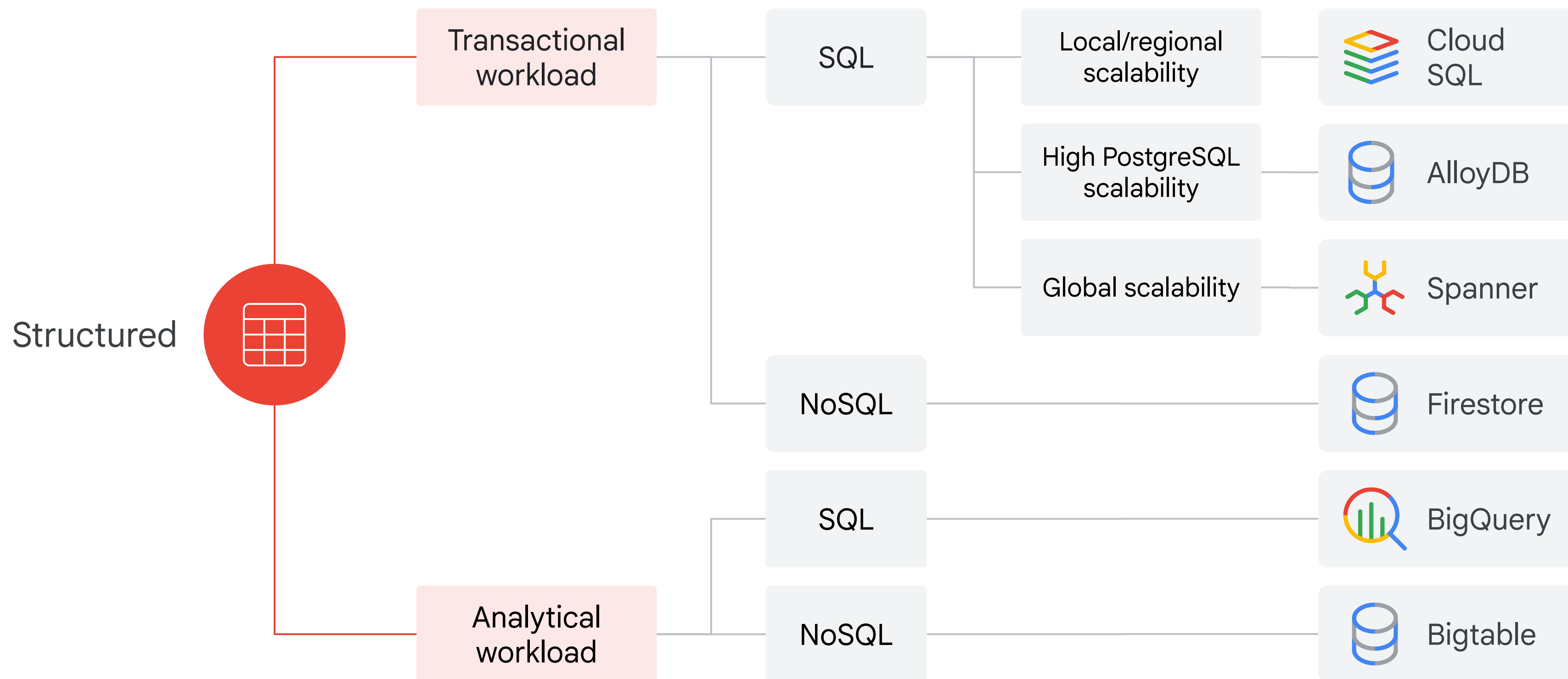
✓ Reliability and scalability

✓ Accessed by HTTP request

✓ Retrieved by object name

✓ Max object size of 5 TB

Options for storing structured data



Data lake versus data warehouse

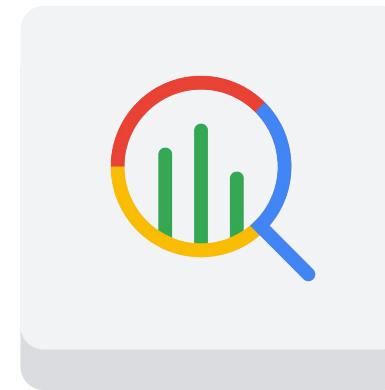
	Data lake	Data warehouse
Data format	Native (unstructured, semi-structured, structured)	Schema (structured or semi-structured)
Data type	Raw	Pre-processed and aggregated from multiple data sources
Purpose	Data science, applications, business decisions	Long-term business analysis
Dependencies	Tools/processes to enable data discovery, governance, security, metadata management	Standalone
Service	 Cloud Storage	 BigQuery

BigQuery is a serverless, fully managed data warehouse

Security on dataset, table, column and row level

Built-in **machine learning** and **geographic information system**

Scalable storage and analytics services



BigQuery

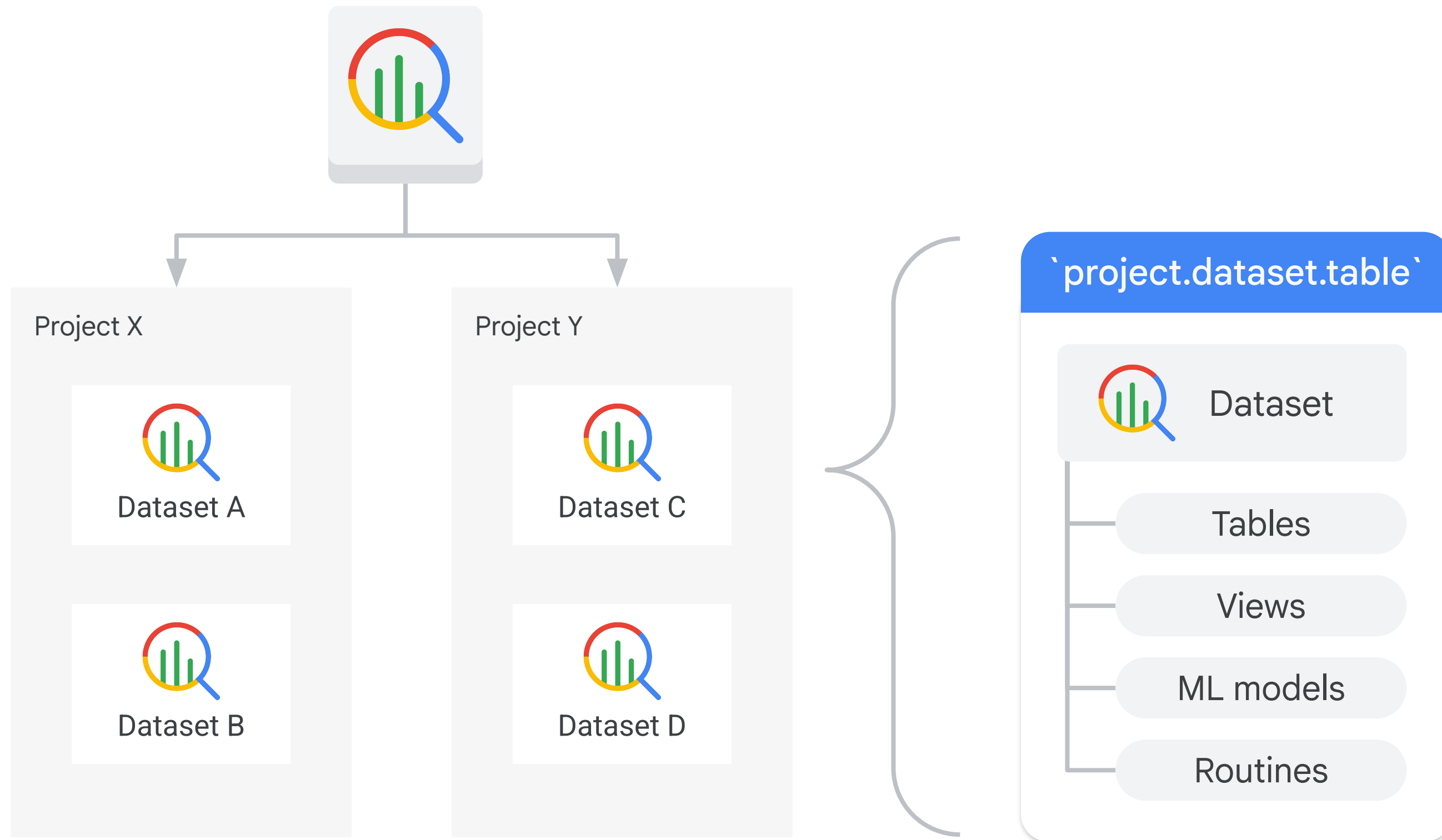
Rich ecosystem for data transformations

Integration with other storage services

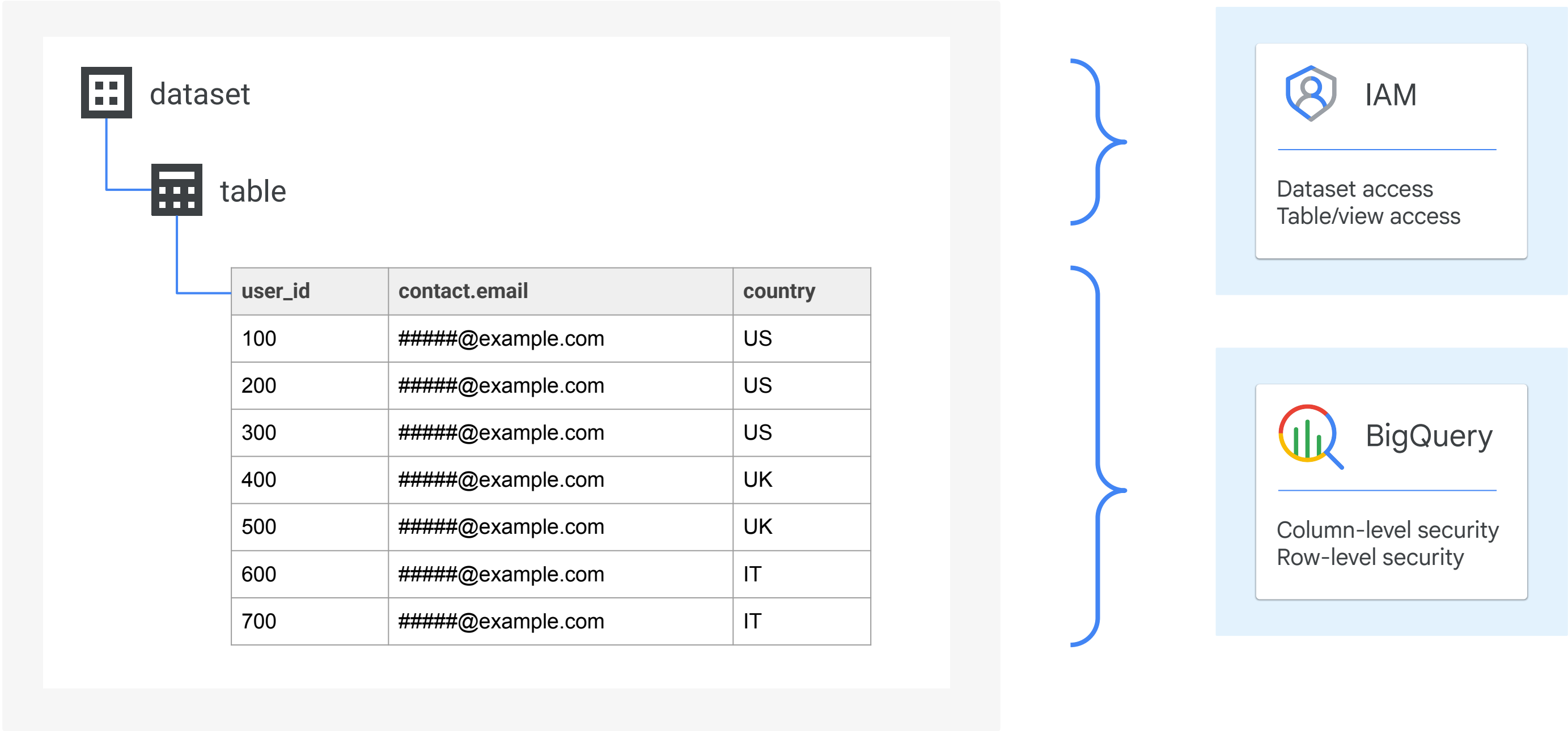
Real-time analytics on streaming data



How BigQuery's resources are organized

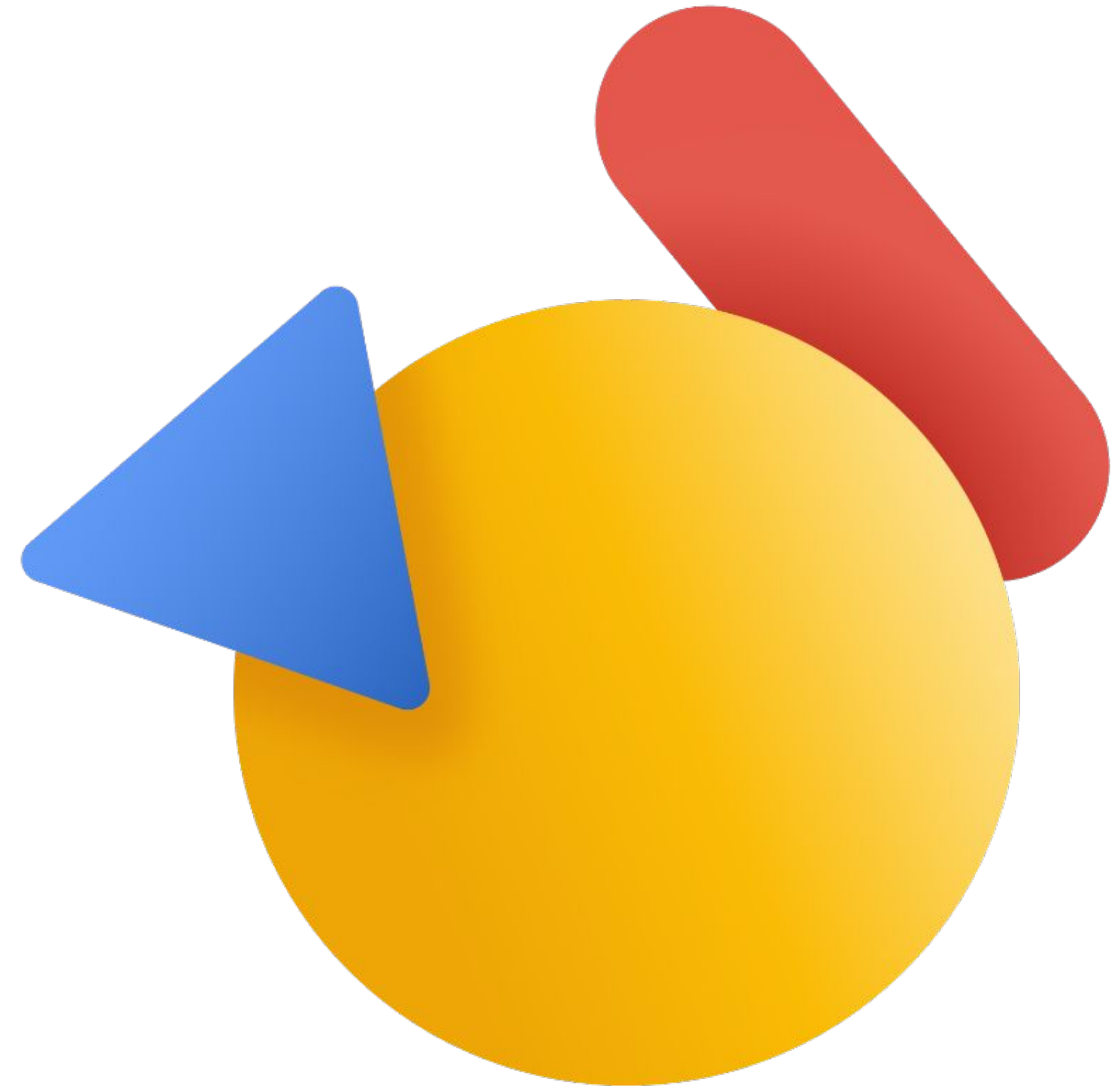


You can secure your BigQuery resources on multiple levels



Demo

Query terabytes of data in seconds



DIN: Exploring projects, datasets and tables

15 minutes · Google Cloud console — your lab project

1. Open the BigQuery console and star the bigquery-public-data and roi-bq-demos projects
2. Create a dataset named class in your lab project — every afternoon loading activity writes into it
3. Browse a public table: preview rows, inspect the schema, find the table details
4. Notice where you are: project → dataset → table, and the region you picked at creation

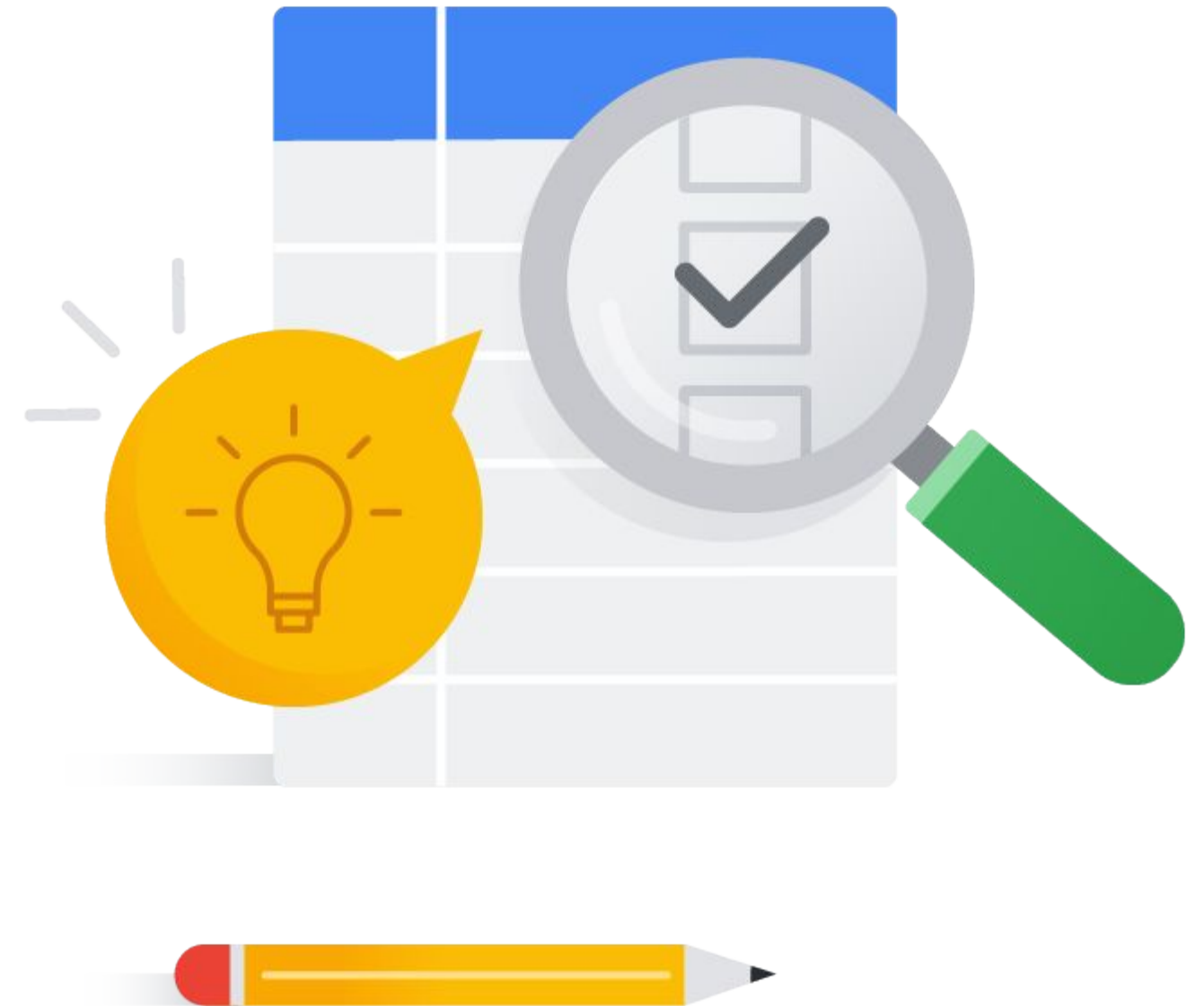
Why it matters — near-zero console experience is the only real gap in this room — this is the 15 minutes that closes it, and the class dataset is required by everything after lunch.

Lab: Loading Data into BigQuery

🕒 30 min

Learning objectives

- Load data into BigQuery from various sources.
- Load data into BigQuery using the CLI and the Google Cloud console.
- Use DDL to create tables.



M1 — What you now have

- ✓ The pipeline shape you know— ingest, transform, store — with Google service names on each box
- ✓ A mental map of the console: projects, datasets, tables, IAM and billing at the project level
- ✓ Your class dataset created— the landing zone for this afternoon's four loading activities
- ✓ Next: killing the nightly extracts— Datastream and change data capture (M2)